

**MBA
USP
ESALQ**

**TREES, NETWORKS AND
ENSEMBLE MODELS**

João F. Serrajordia R. de Mello

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Introduction

João Fernando Fernando Serrajordia Rocha de Mello –
(Juka)

Professional journey

Credit modeling in large banks

Telecom

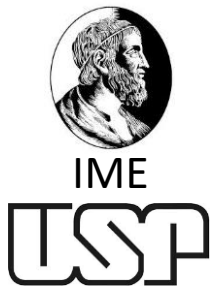
Development of models / Validation of models

Teaching in data science

Consultancy in data science

Executive *outsourcing*

ACADEMIC



BACHAREL EM ESTATÍSTICA

MESTRE EM ESTATÍSTICA



What will we learn?

- Decision Trees
- Ensemble
 - Bagging (Random Forest)
 - Boosting (Gradient Boosting)
 - Principles of cross-validation
- Introduction to Neural Networks



Today's agenda

- Conception
- Decision Trees
 - Investments
 - What is it, how does it run, etc...
 - Operational details (base etc.)
 - Algorithm
 - How to evaluate it?
 - Common problems





Let's discuss...

What is machine learning
?

What is Machine Learning?

- Is it a branch of Artificial Intelligence?
- Calm down... What is Artificial Intelligence after all?

'Mundane' definition



Common sense: Artificial intelligence is the area that studies the implementation of activities performed by machines, which were previously performed by humans.



Oxford Dictionary:

“the theory and development of computer systems able to perform tasks that no other machine can do.”

“the theory and development of computational systems that perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision making and translation between languages.”



Definition of MIT

- “Artificial intelligence is the ability of computers to imitate human cognitive functions, such as learning and troubleshooting. Through AI, a computer system uses mathematics and logic to simulate the reasoning that people use to learn from new information and make decisions.”

<https://mitsloan.mit.edu/>

Academic definition

- “We define AI as the study of agents that receive percepts from the environment and perform actions.”
Artificial Intelligence: a modern approach, by Stuart Russell and Peter Norvig

“We define AI as the study of agents that receive percepts from the environment and perform actions.”





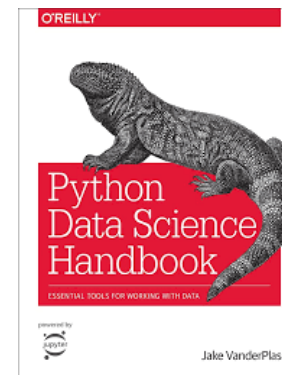
Let's discuss it...

And what is machine learning...?

Machine learning

- Jake Van Der Plas:
 - It helps to think about machine learning as a way of building models on data.
 - It involves the construction of mathematical models to understand data.
 - The “learning” is in the possibility of obtaining regulable parameters that can be adequate to the observed data.

<https://jakevdp.github.io/PythonDataScienceHandbook/05.01-what-is-machine-learning.html>



You will need...



Preparations

- Open R
- Import libraries
- Electronic sheet
- Something to take your notes



Decision Trees

Decision Trees:

Where do they live?

What are they?

What do they eat?

Natural Predators

Investments



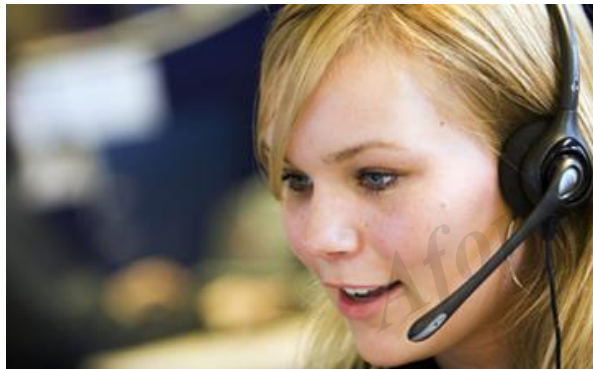
What is the efficacy of a vaccine?



Will the customer pay the loan?



How much oil is in the well?



Will the customer buy my product?

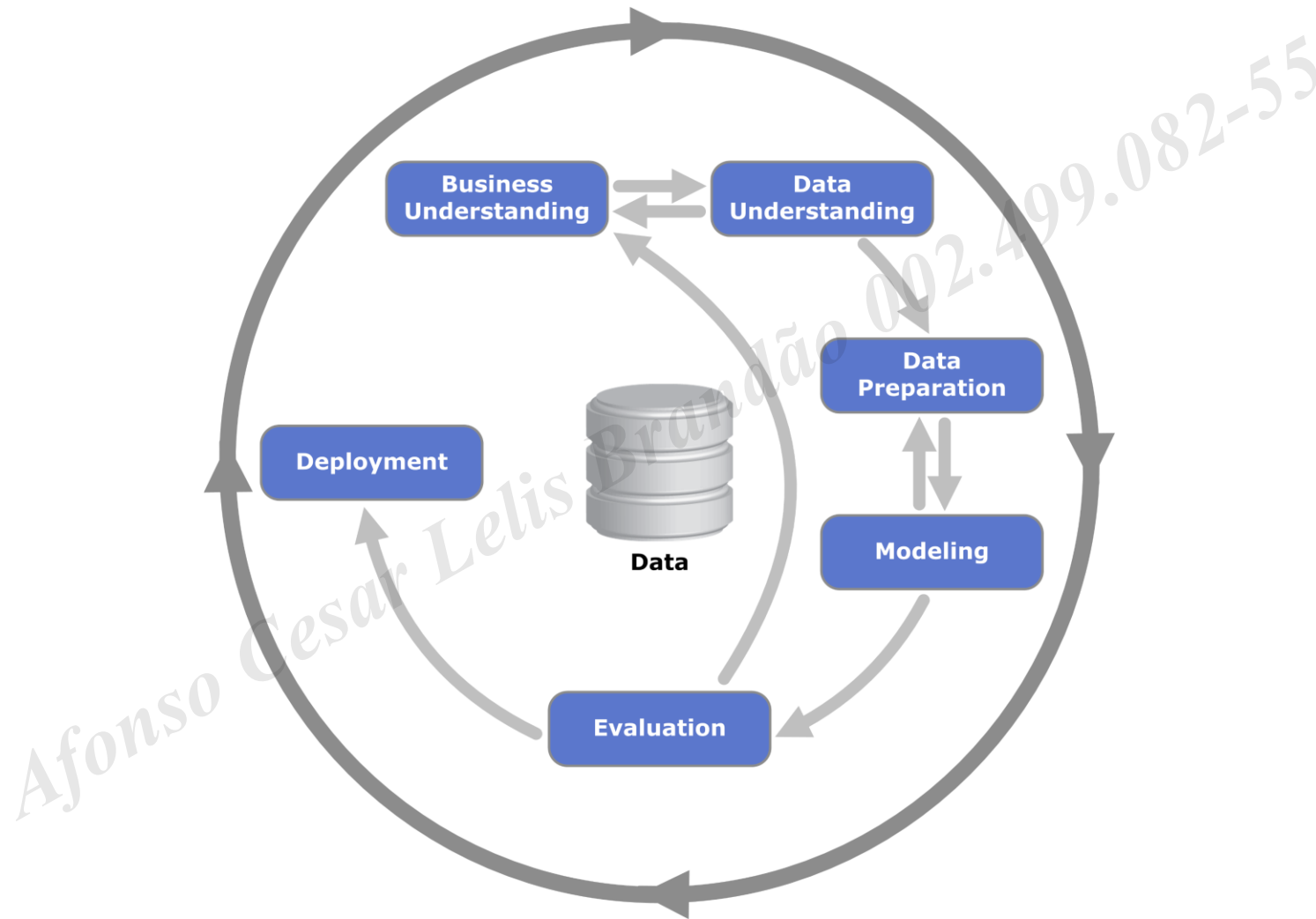


What is the person doing?



How green is this vehicle?

CRISP-DM



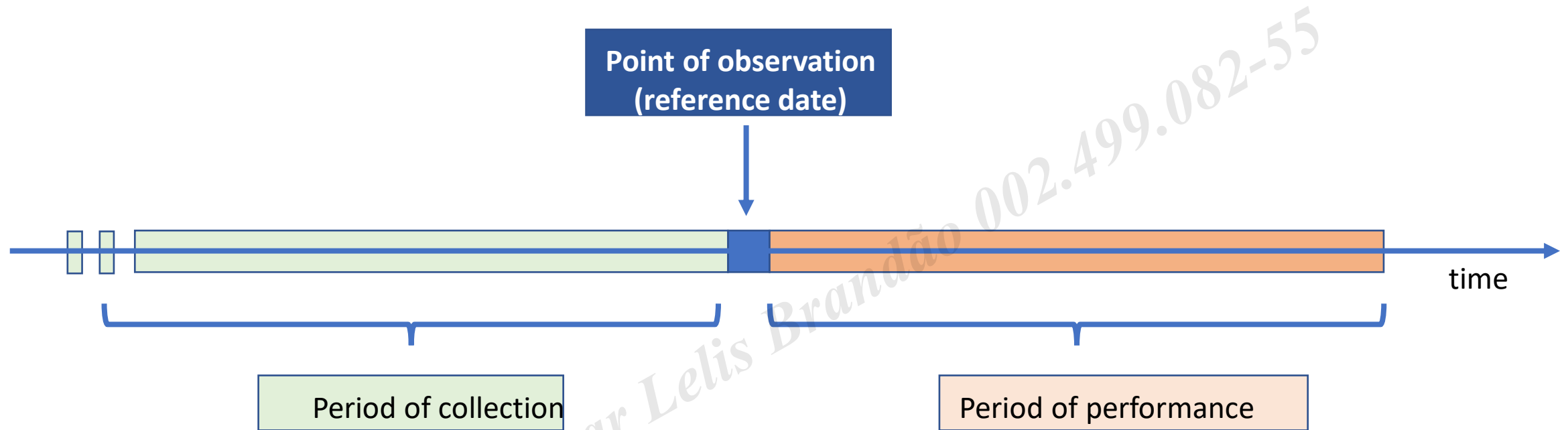
Source: <https://www.the-modeling-agency.com/crisp-dm.pdf>



Predictive models

How is this?

Cohort



Example of drawing a sample for predictive model

Structure of the database

Each column is a variable.

Each line
represents a
observation

ID_cliente	Idade	Renda	Tempo de emprego	Negativado_Bureau	...	Inadimplente_em_12m
2914901	60	75579	10	0	...	0
5003858	25	33495	2	0	...	1
8309347	51	59908	9	0	...	1
4637814	46	53470	10	0	...	0
7203409	65	69822	1	0	...	1

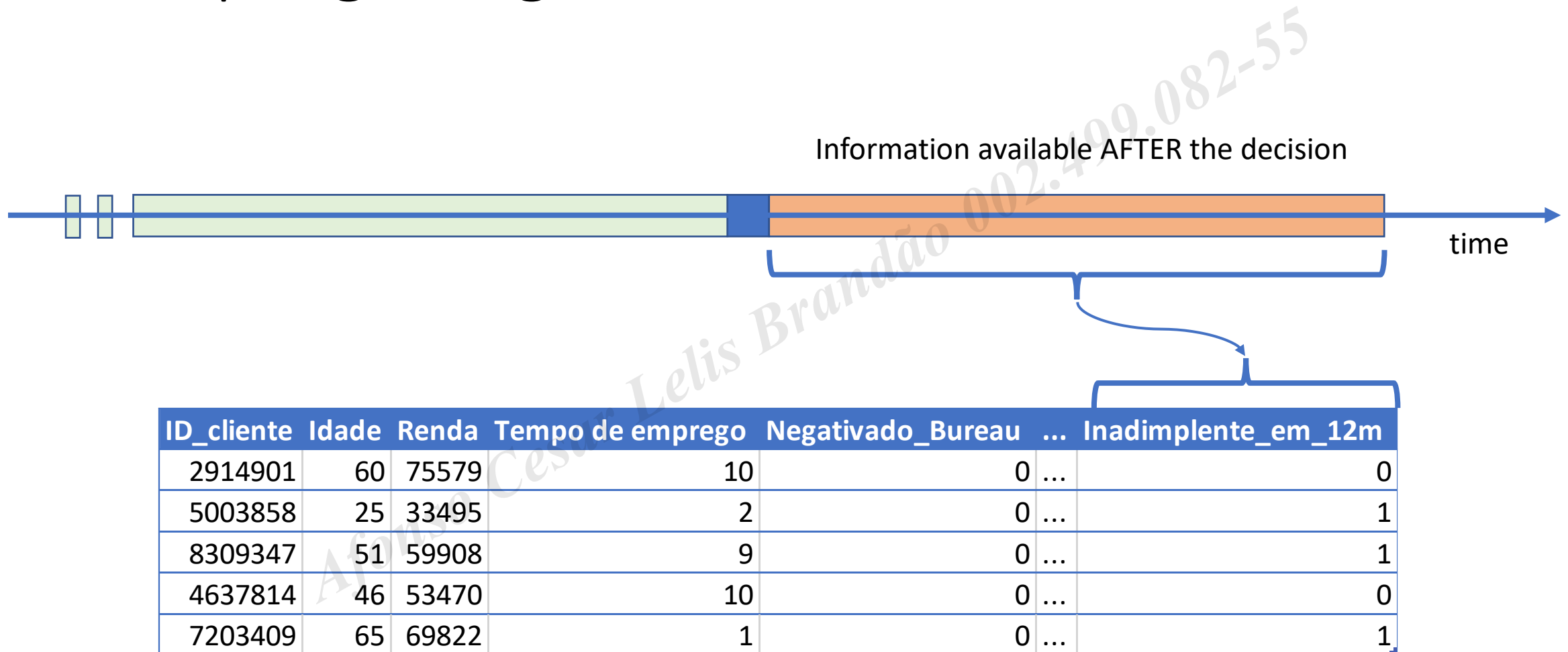
Sampling design

Information available BEFORE the decision.

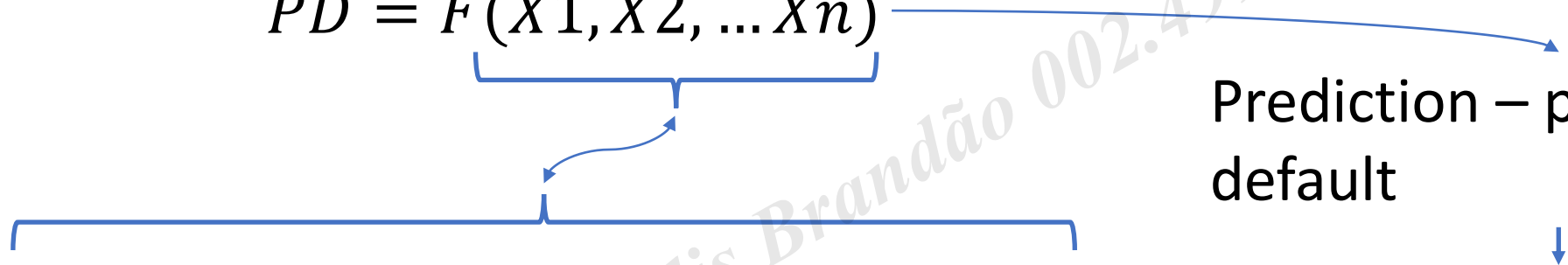


ID_cliente	Idade	Renda	Tempo de emprego	Negativado_Bureau	...	Inadimplente_em_12m
2914901	60	75579	10	0	...	0
5003858	25	33495	2	0	...	1
8309347	51	59908	9	0	...	1
4637814	46	53470	10	0	...	0
7203409	65	69822	1	0	...	1

Sampling design



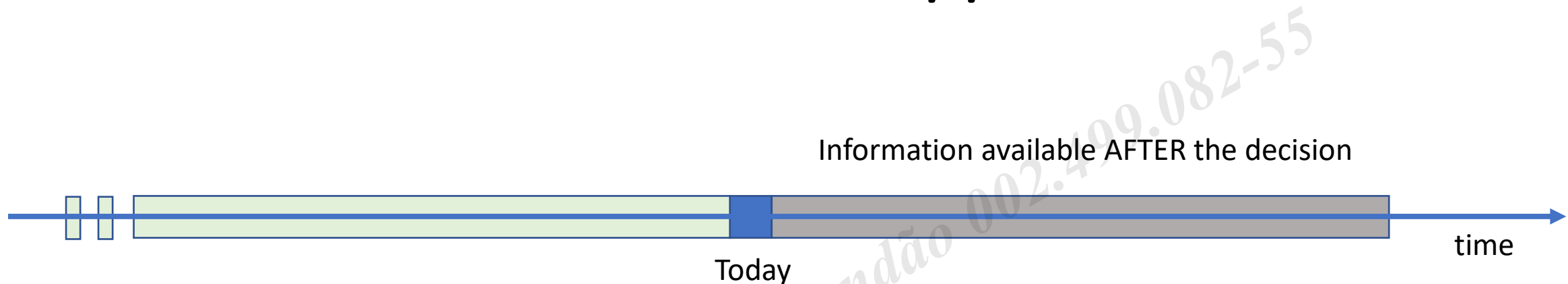
Prediction of the model - *training*

$$PD = F(X1, X2, \dots Xn)$$


Prediction – probability of default

ID_cliente	Idade	Renda	Tempo de emprego	Negativado_Bureau	...	Inadimplente_em_12m	Prob(Inad_12)
6046023	63	44095	7	1	...	1	64,2%
6953168	45	47550	8	1	...	1	63,3%
8529261	58	89313	7	1	...	1	71,3%
1281888	42	58532	7	0	...	0	15,3%
4540759	37	80840	2	0	...	1	86,7%

Prediction of the model - *application*



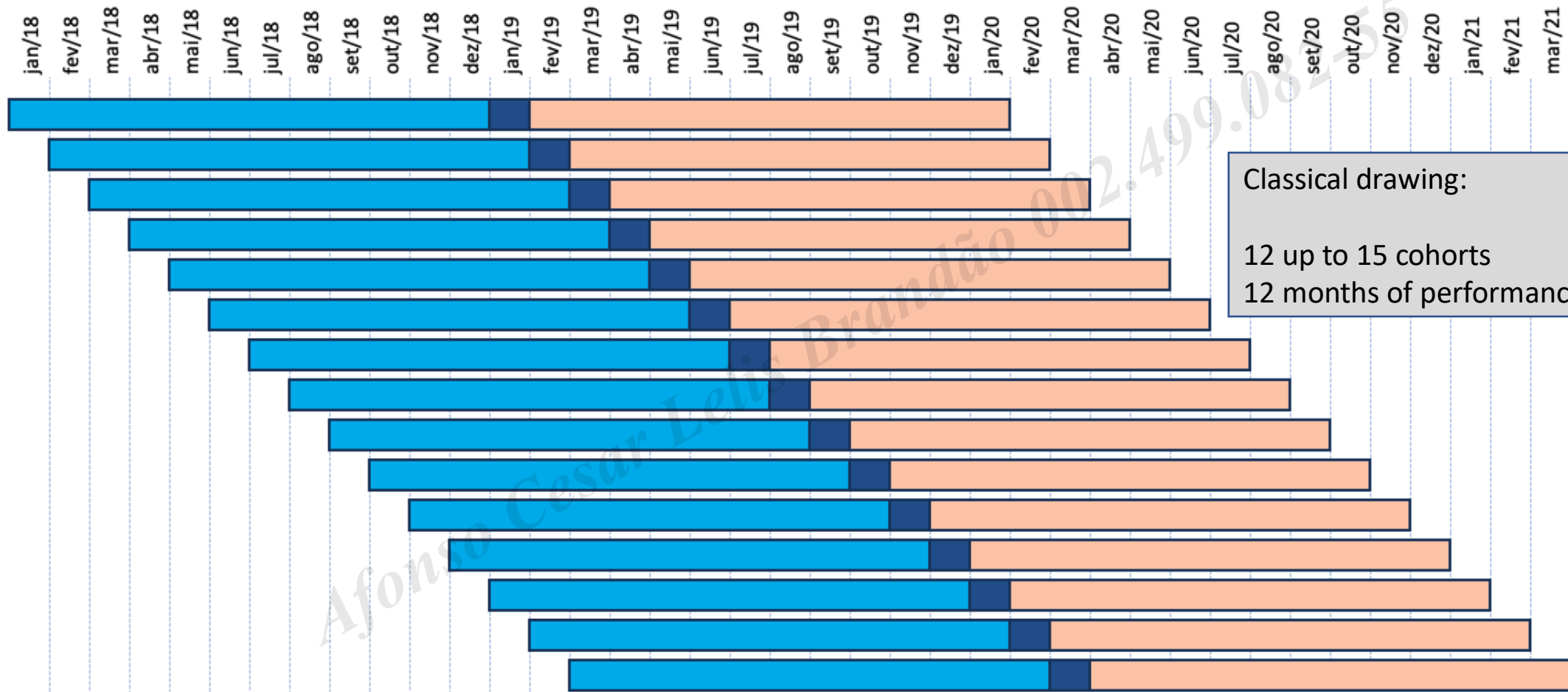
ID_cliente	Idade	Renda	Tempo de emprego	Negativado_Bureau	...	Inadimplente_em_12m	Prob(Inad_12)
2064492	57	78774	10	0	...		26,4%
8858083	63	99559	4	1	...		39,4%
6937124	41	49791	2	0	...		93,6%
7352807	42	15314	8	0	...		34,1%
5088775	31	73199	4	0	...		97,4%

In the application, we do not have the marking of default but we have the probability given by the model.
This is the PREDICTION of the model.

Classical design

- What if the month of observation suffers some seasonal effect as the end of the year etc?
- For this, it is possible to observe 12 months of different reference (called “cohort”)

Model Drawing



Classical drawing:

12 up to 15 cohorts

12 months of performance

Decision Trees:

Where do they live?

What are they?

What do they eat?

Natural Predators

Algorithms classification

Machine Learning Paradigm

- Decision Trees
- Bagging
- Boosting
- K-NN
- Neural Networks
- Support Vector Machines

Statistical Paradigm

- Regression
- GLM
- GLMM
- ANOVA

We are here!

Machine learning vs statistics

- Machine Learning

- It only searches for patterns in the data
- It searches for making them generalizable
- It measures the error in a pragmatic way



- Statistical Paradigm

- It supposes probabilistic structure
- It evaluates the assumptions
- **It makes inference**



Algorithms classification

Supervised

- Regression
- GLM
- GLMM
- Support vector machines
- Naive Bayes
- K-nearest neighbors
- Neural Networks
- Decision Trees



Unsupervised

- K-Means
- Hierarchical methods
- Gaussian Mixture
- DBScan
- Mini-Batch-K-Means



We are here!

Algorithms classification



Continuous response

- Regression
- GLM
- GLMM
- Support vector machines
- K-nearest neighbors
- Neural Networks
- Regression Trees



Discrete response

- Logistic Regression
- Classification trees
- Neural Networks
- GLM
- GLMM

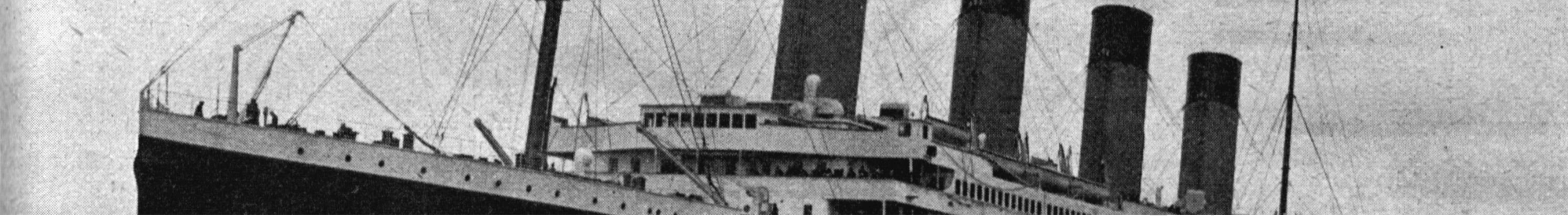
We are here!



Our problem: classify survivors

Image: [https://commons.wikimedia.org/wiki/File:Sea Trials of RMS Titanic, 2nd of April 1912.jpg](https://commons.wikimedia.org/wiki/File:Sea_Trials_of_RMS_Titanic,_2nd_of_April_1912.jpg)

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Reflections on the database

Population

- ~ 2,200 people
- ~ 1,300 passengers
- More than 1,500 deaths

Sample

- 891 people
- 549 non-survivors
- 342 survivors



Objectives of the algorithm of Machine Learning

- To classify the response variable as well as possible
 - ... Through segmentations
 - ... Using explanatory variables
- To obtain insights
 - ... From relations between the response variable and the explanatory variable
 - ... To explore interactions

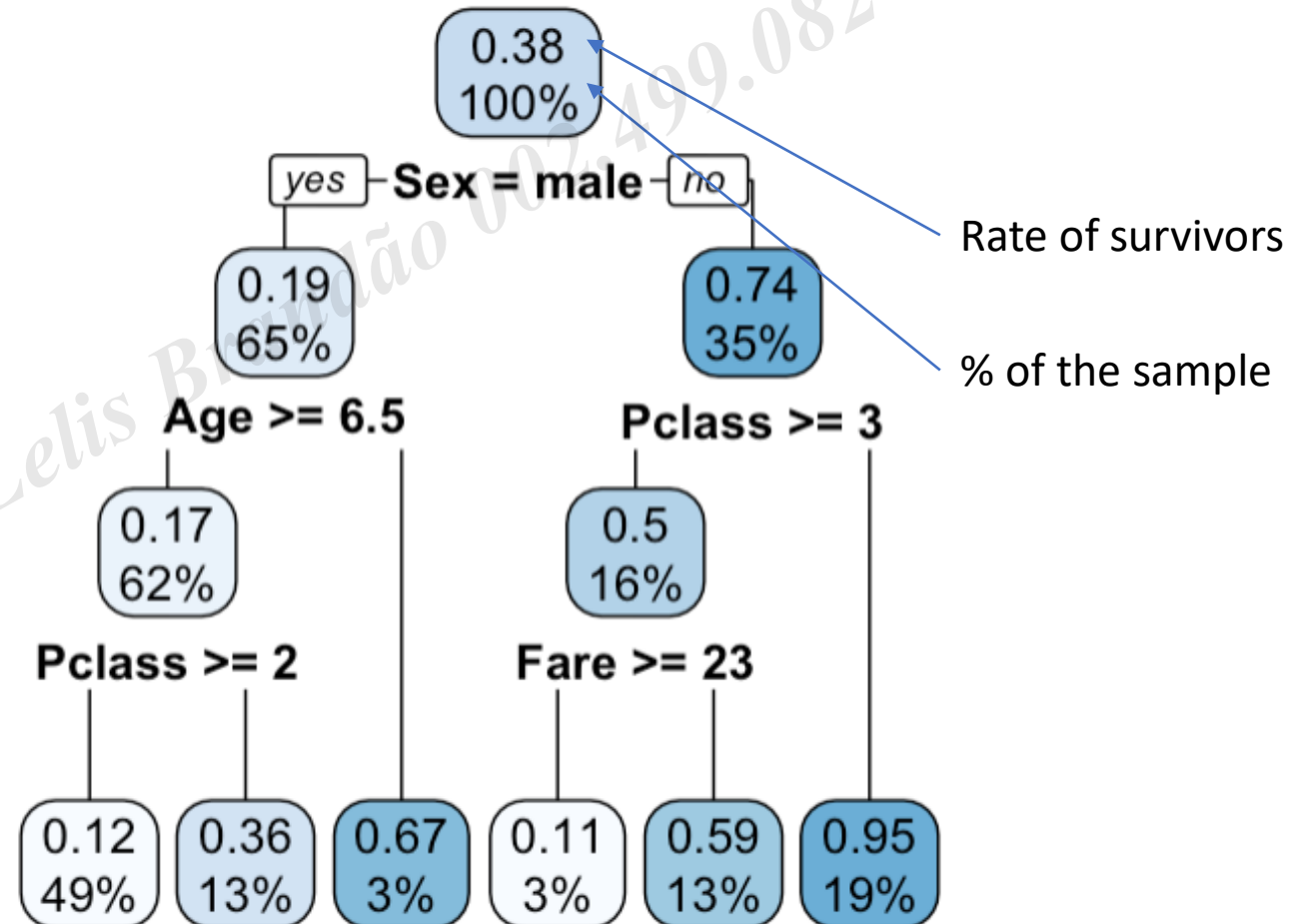


OMML1_script01-Primeiro_contato_com_arvores.R

What is a decision tree?

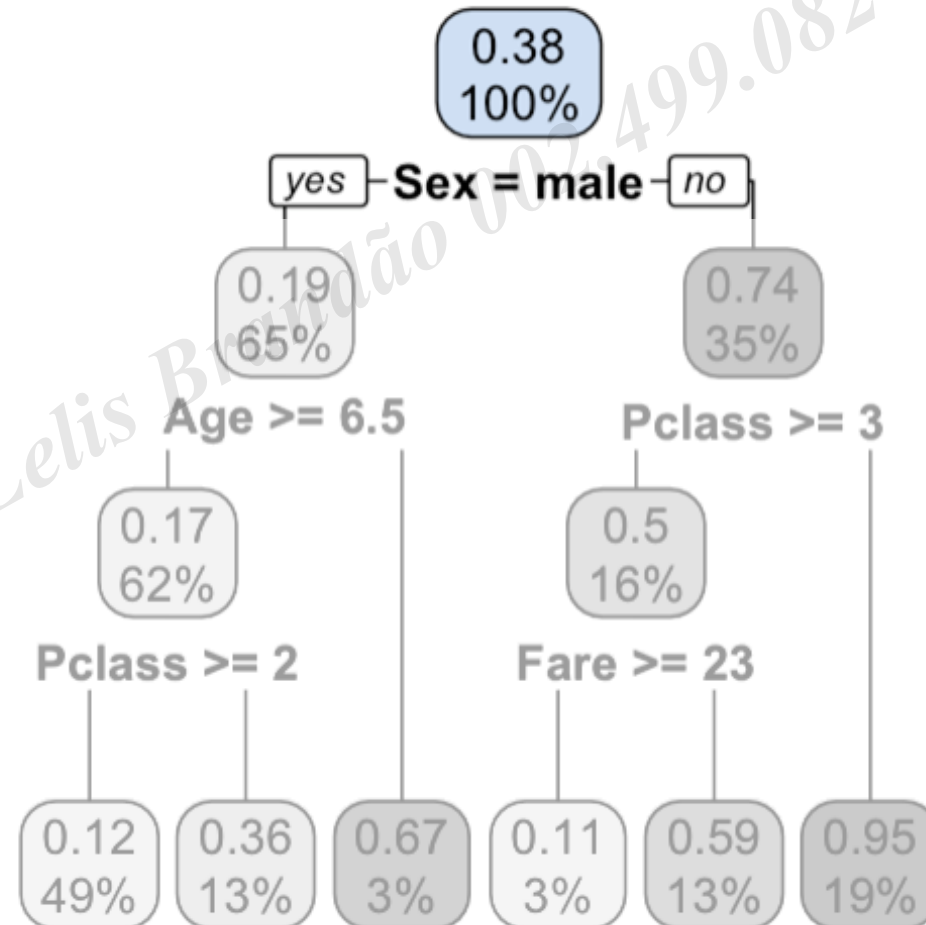
Decision tree is:

A sequence of binary segmentations
That aims the homogeneity of the response variable



What is a decision tree?

Initially we have 891 passengers of which
342 survived (38%)
549 did not survive

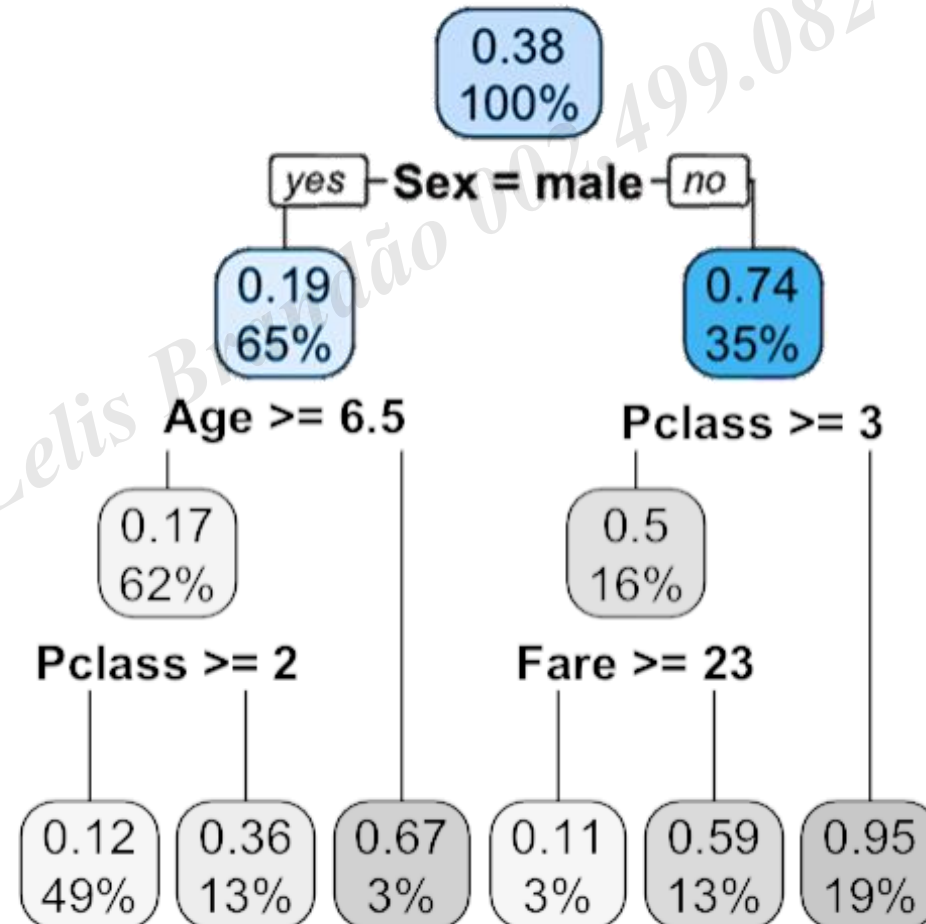


What is a decision tree?

We can segment from the 891:

577 men (65%) of which
109 survived (19%)
468 did not survive

314 women (35%) of which
233 survived (74%)
81 did not survive



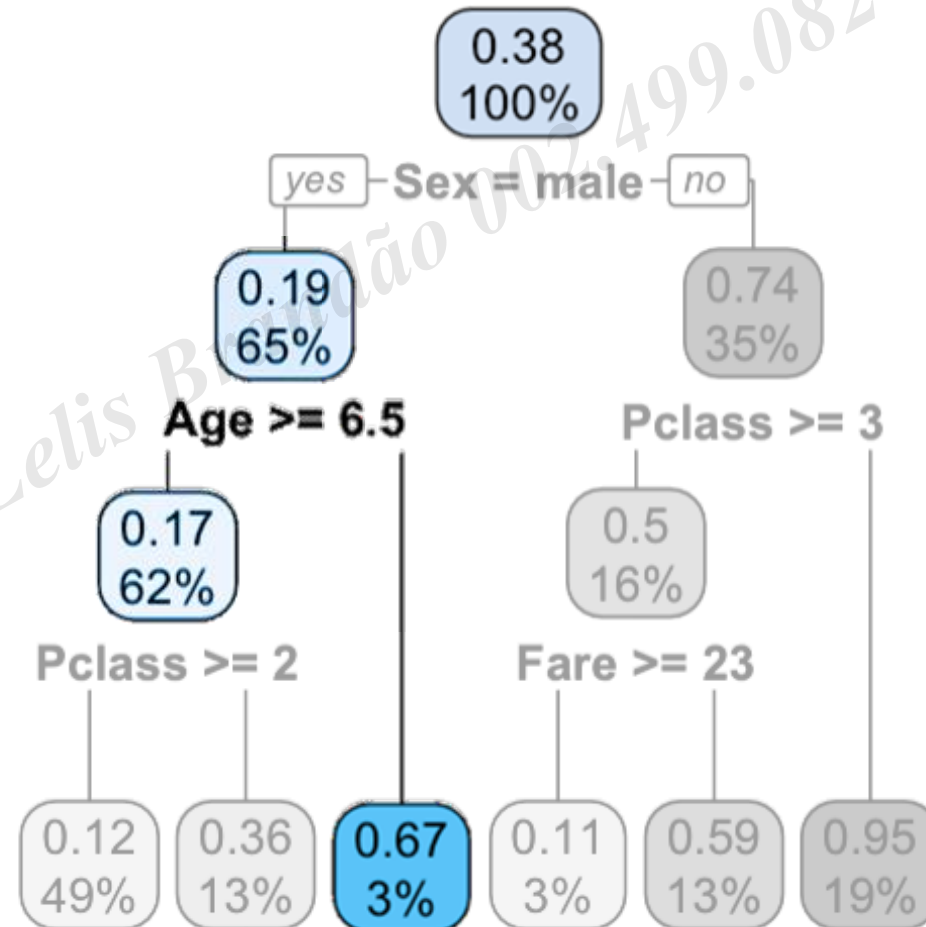
What is a decision tree?

We can segment from the 891:

577 men that we segment in:

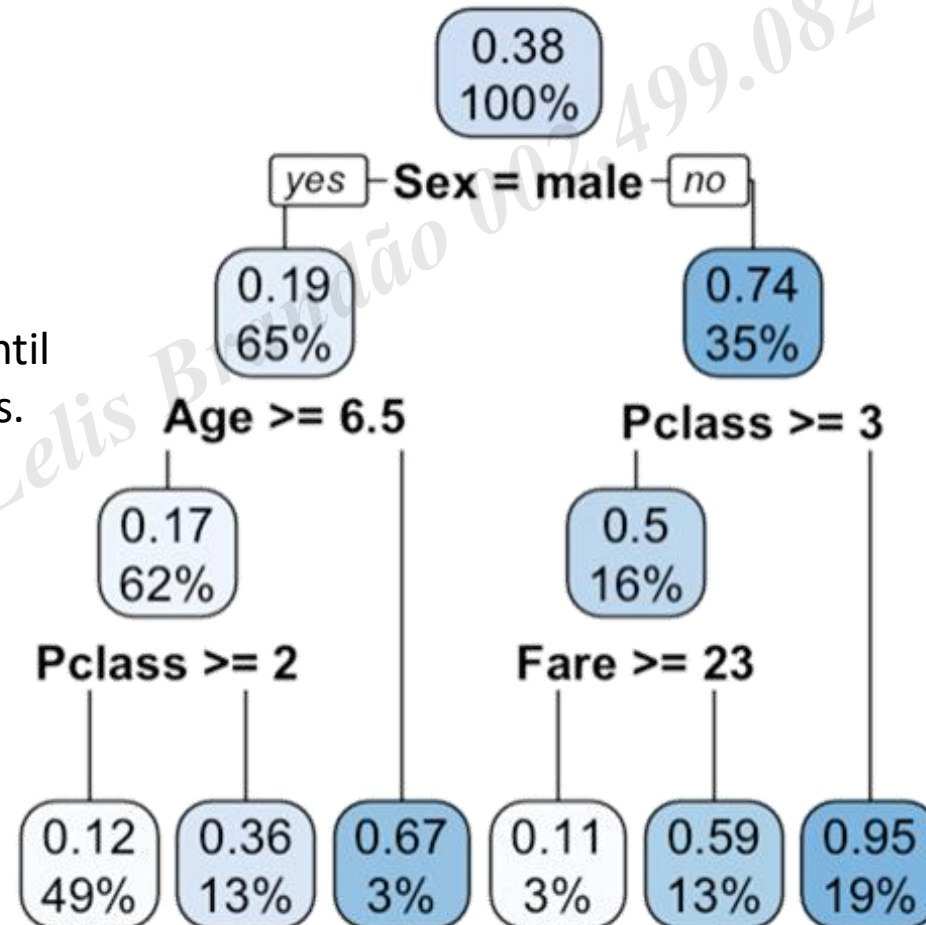
24 kids (< 6, 5 years) of which
16 survived (67%)
8 did not survive

533 adults ($\geq 6, 5$ years) of which
93 survived (17%)
553 did not survive



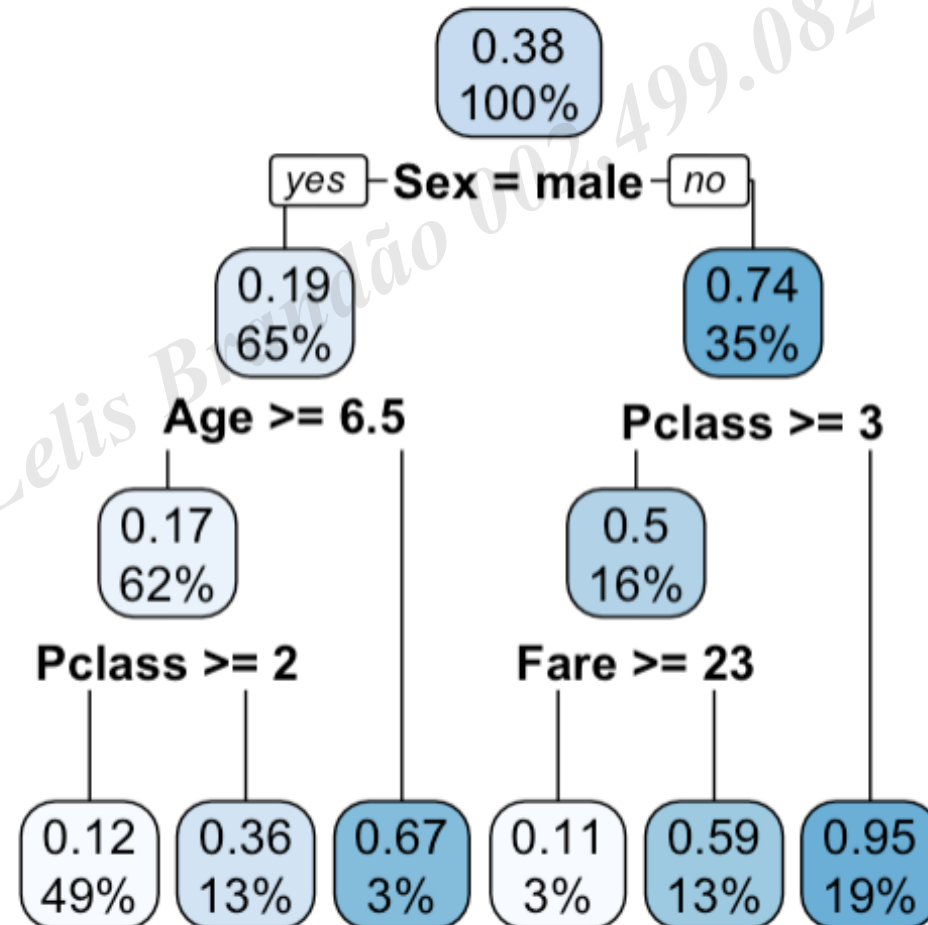
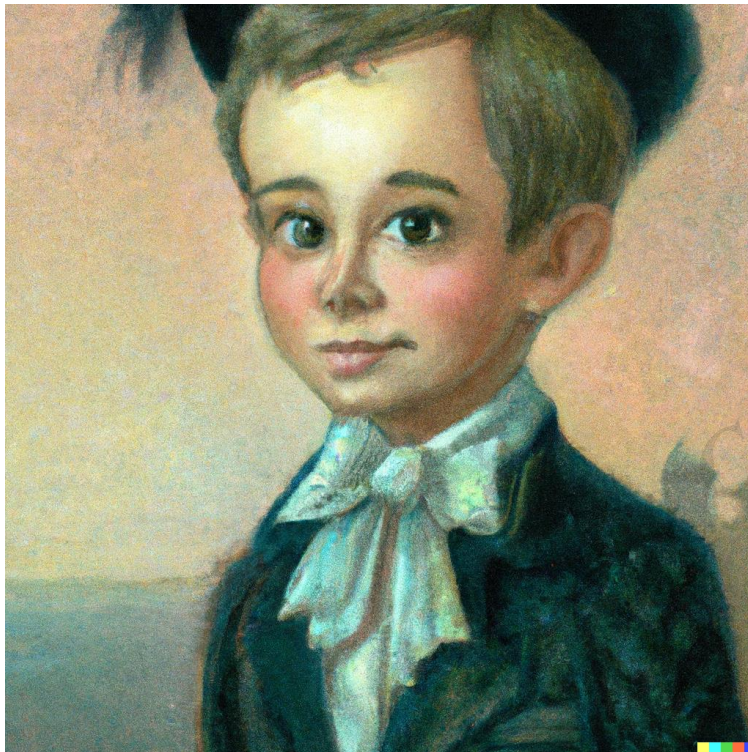
What is a decision tree?

And, then, we continue to "split" the sample until "it is not worth it anymore" to make more splits.



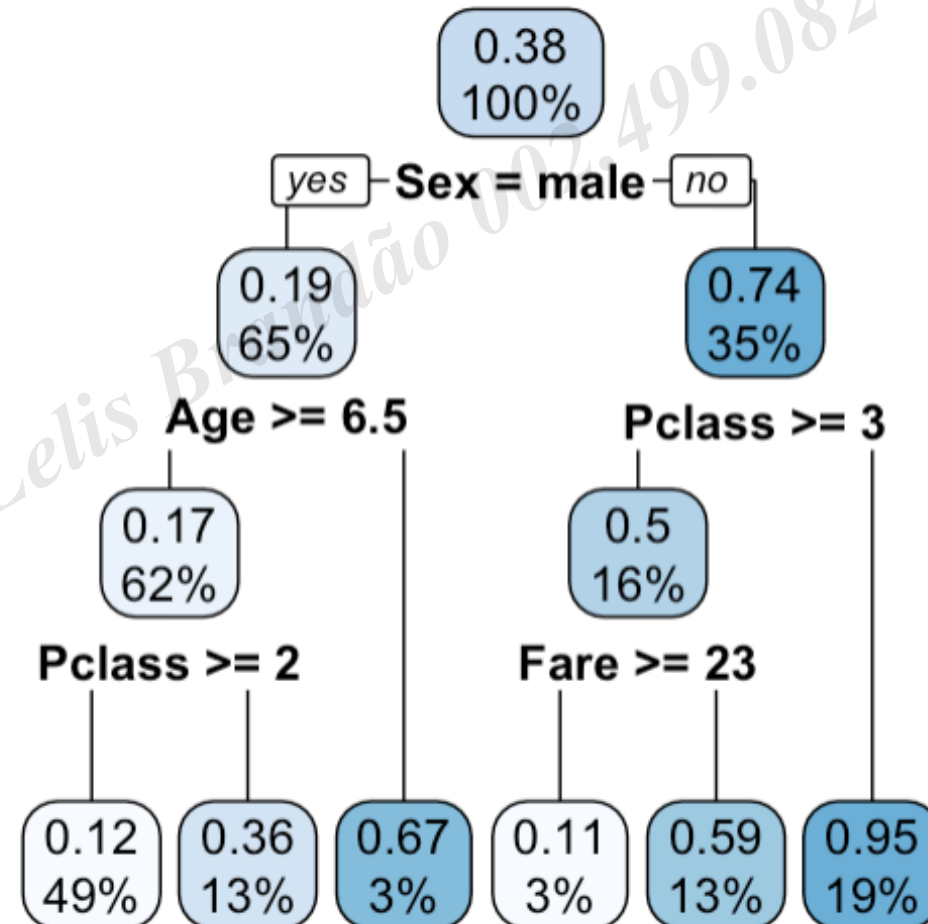
Shall we try??

Let's classify this boy:



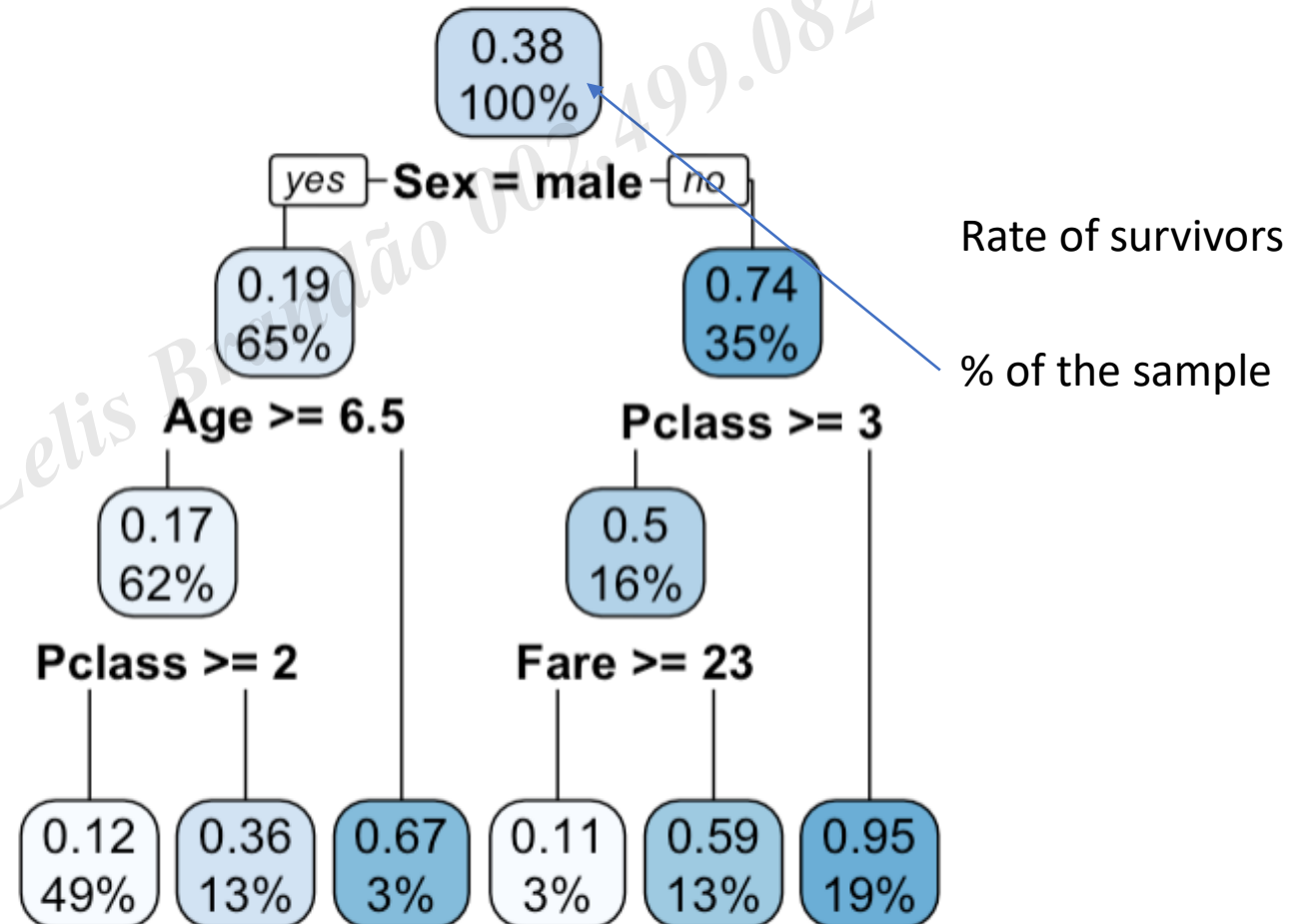
Shall we try??

Let's classify this poor man:



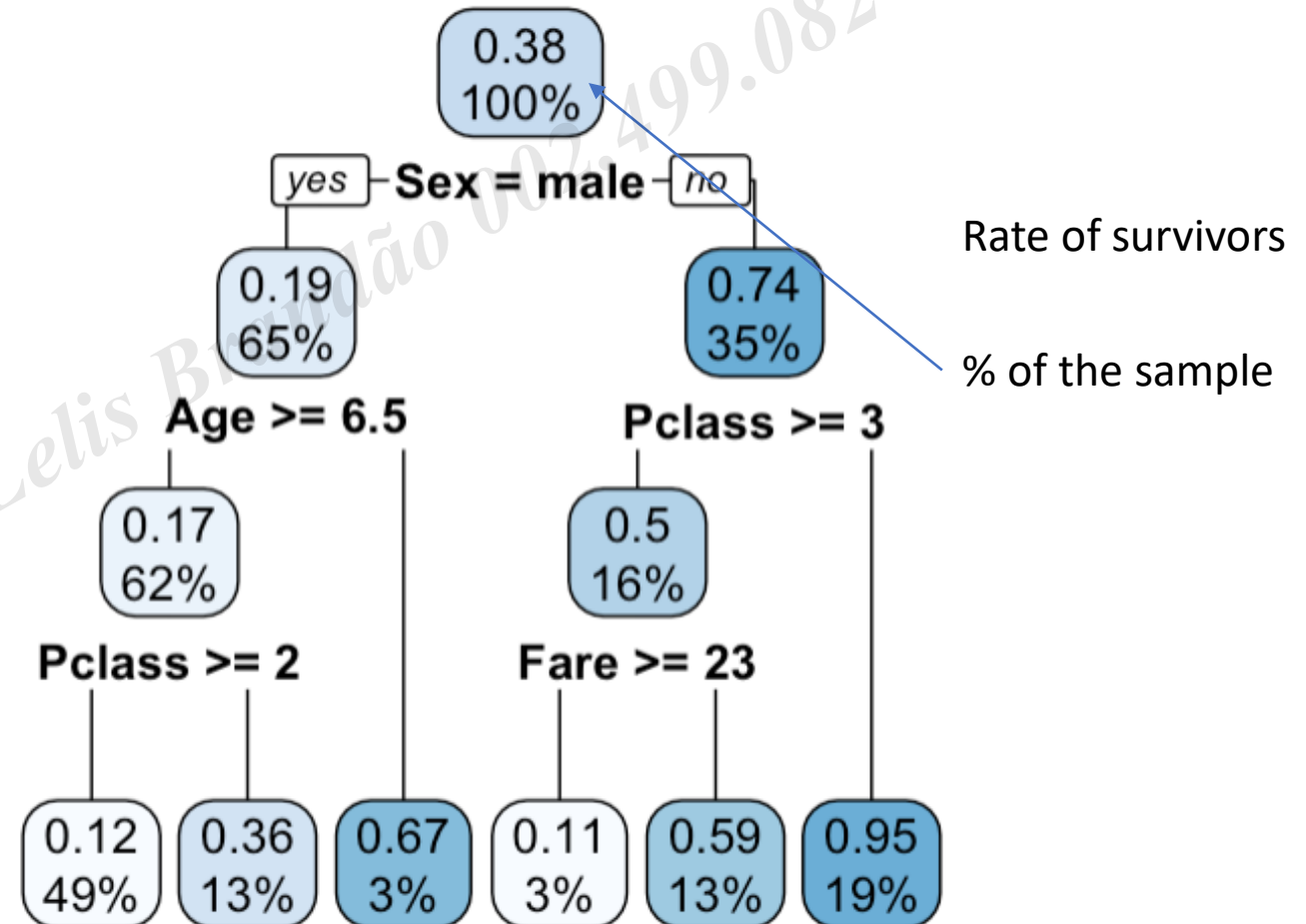
Shall we try??

Let's classify this rich man:



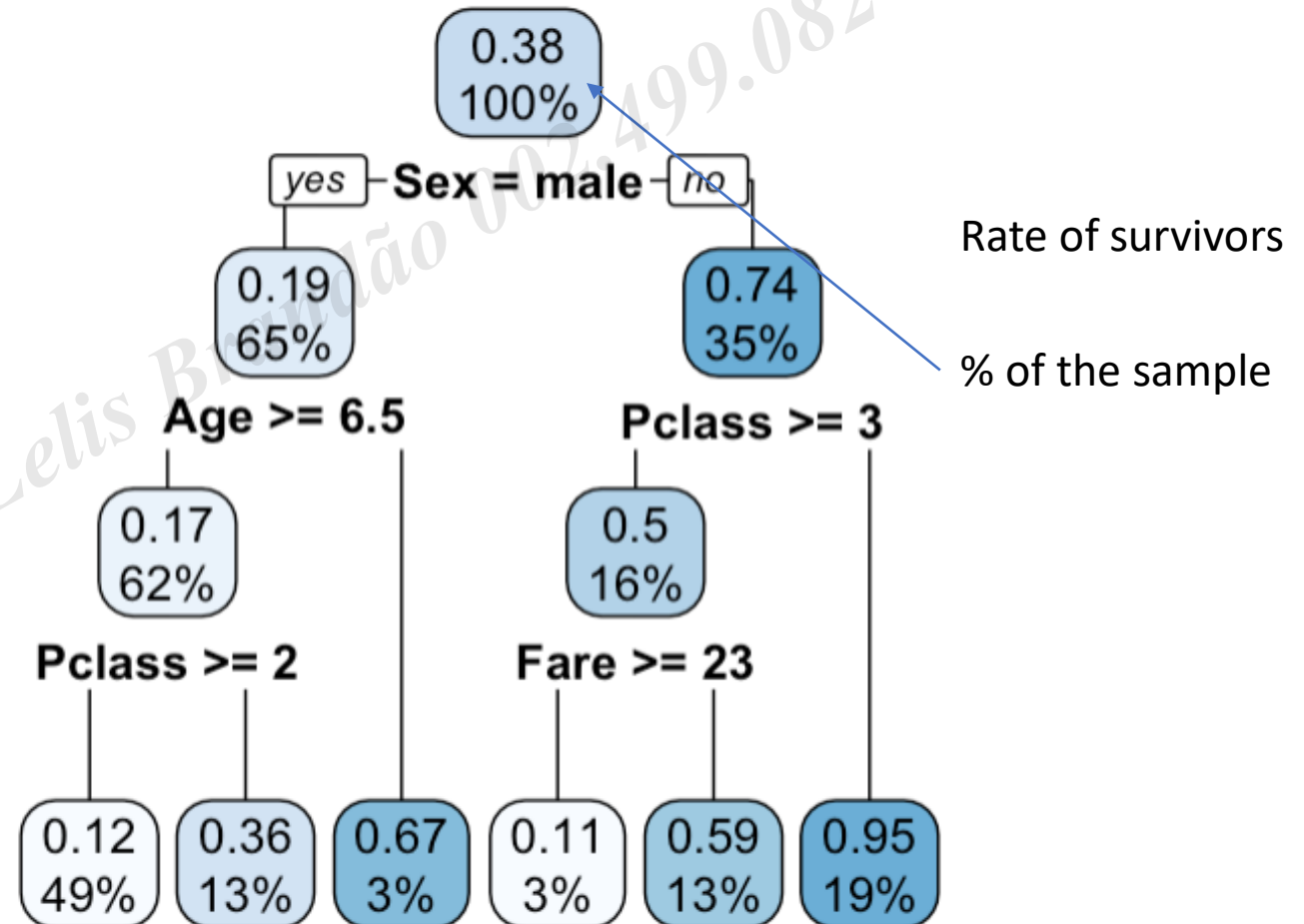
Shall we try??

Let's classify this rich lady:



Shall we try??

Let's classify this poor girl:



Decision Trees:

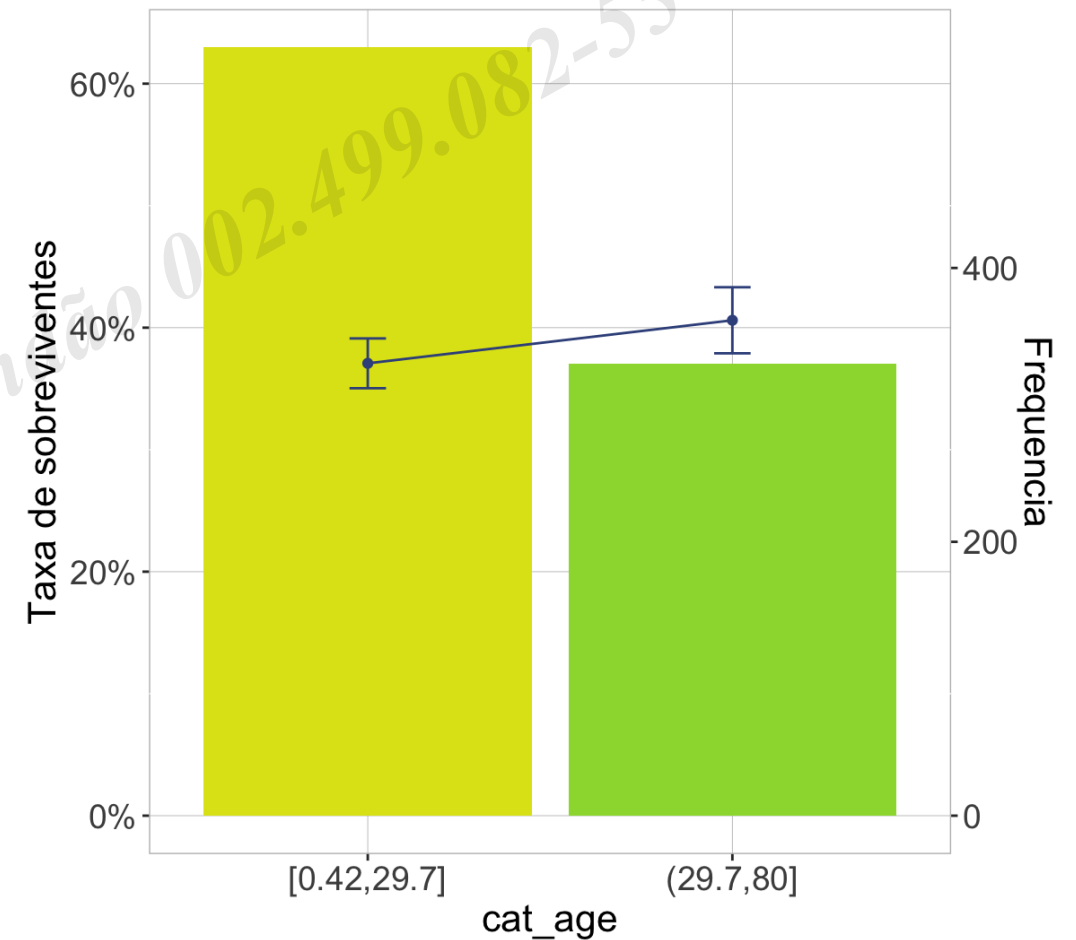
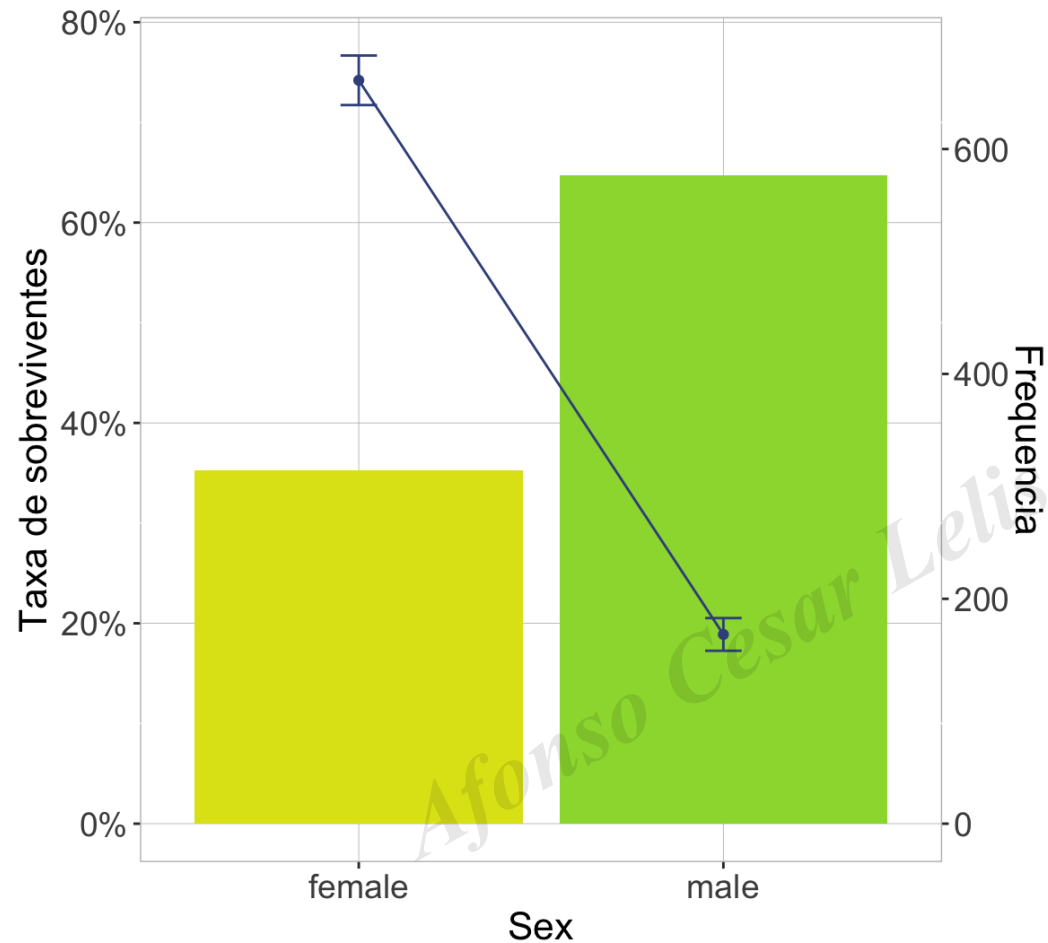
Where do they live?

What are they?

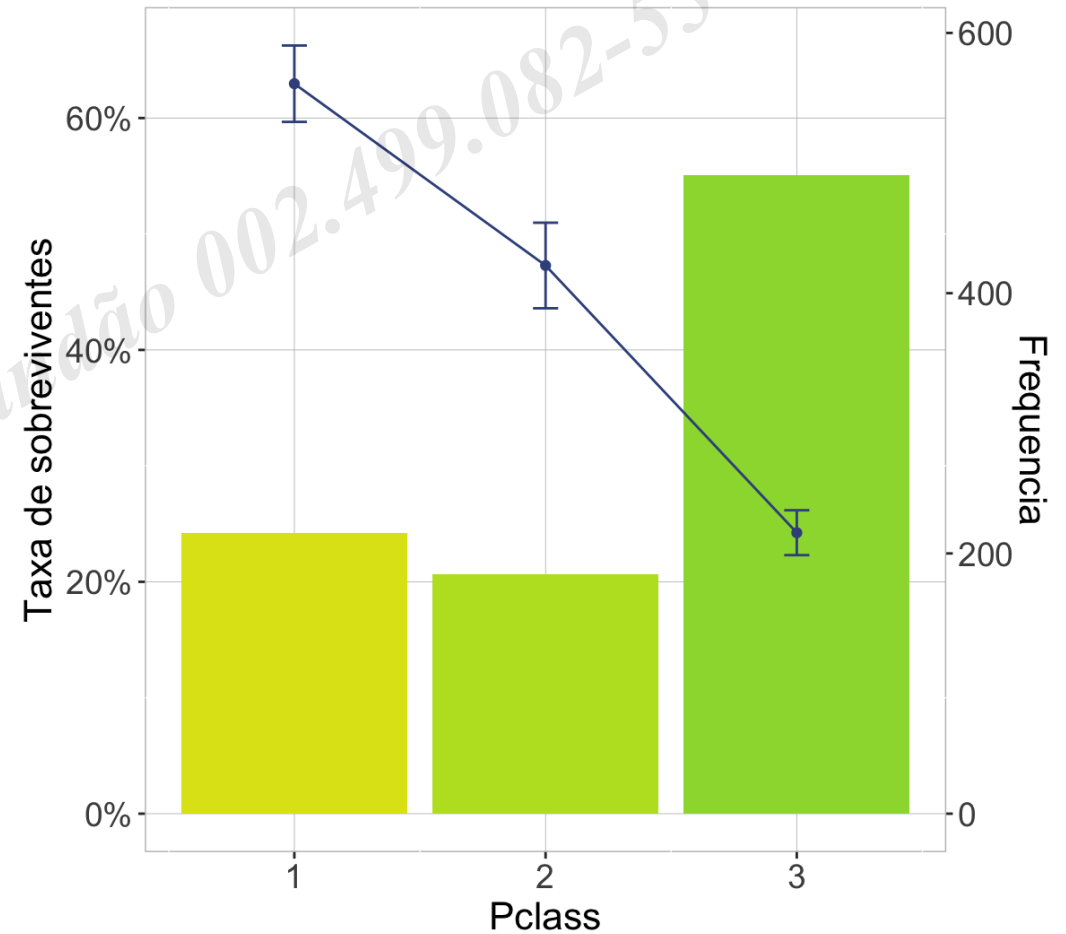
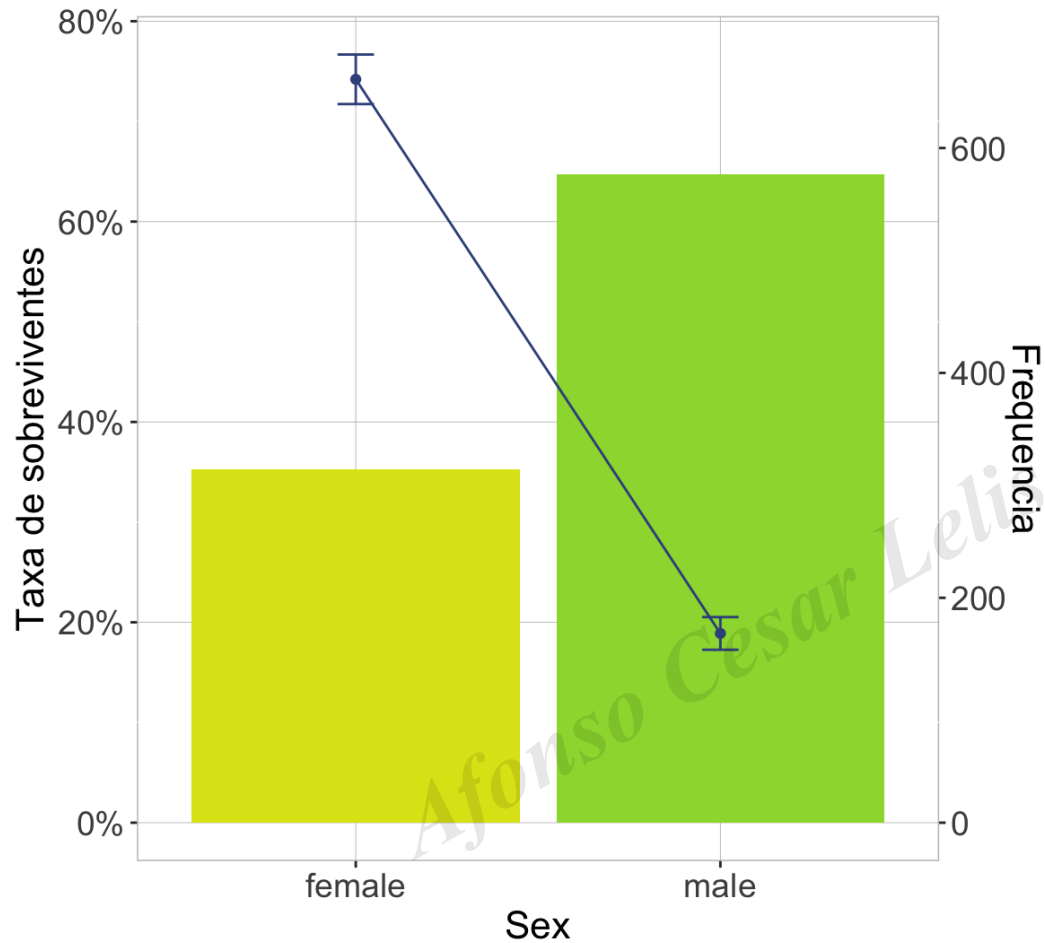
What do they eat?

Natural Predators

Which one is 'better'?



And now?



Measure of “information”

- We need metrics to help decide which variable adds more information.
- Let's say that if all the observations are identical in a group, this group has total “purity”.
- The more heterogeneous the group, the greater the “impurity” will be.
- Let's define "impurity" quantitatively

Definitions of impurity

- Gini
- Shannon Entropy

How does the tree select the best split?
Using a metrics of "impurity"

Gini's index

$$I_g(p) = 1 - \sum_{i=1}^J p_i^2$$

- Maximum impurity with uniform distribution
- Minimum impurity in the total concentration

Entropy

$$H = - \sum_{i=1}^J p_i \log_2(p_i)$$

Information gain:

$$GI(T, a) = H(T) - H(T|a)$$

- Maximum impurity with uniform distribution
- Minimum impurity in the total concentration

Basic Algorithm

1. Seek the best binary rule for each variable.
2. Seek to apply the best segmentation among all variables
3. Recursively, for each sheet, repeat the steps 1 and 2 until a stopping rule is reached.

Interactive web implementation:

<https://rawgit.com/longhowlam/titanicTree/master/tree.html>

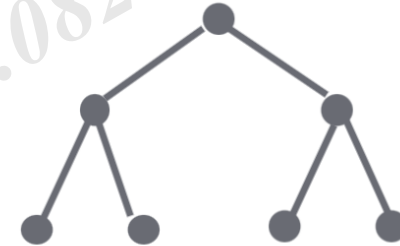
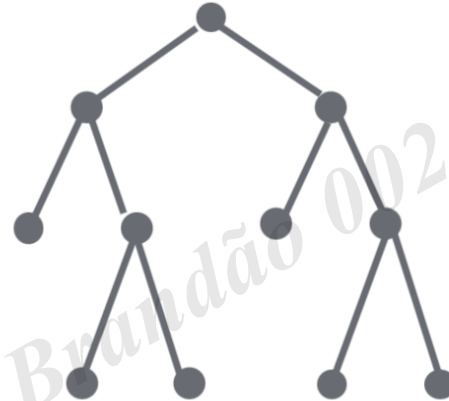
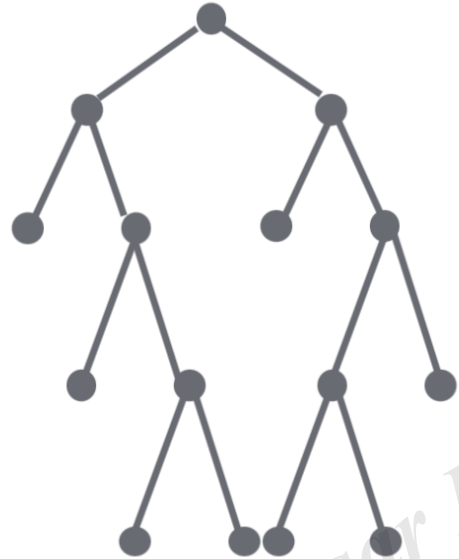


Hyperparameters

They are parameters that control the algorithm as:

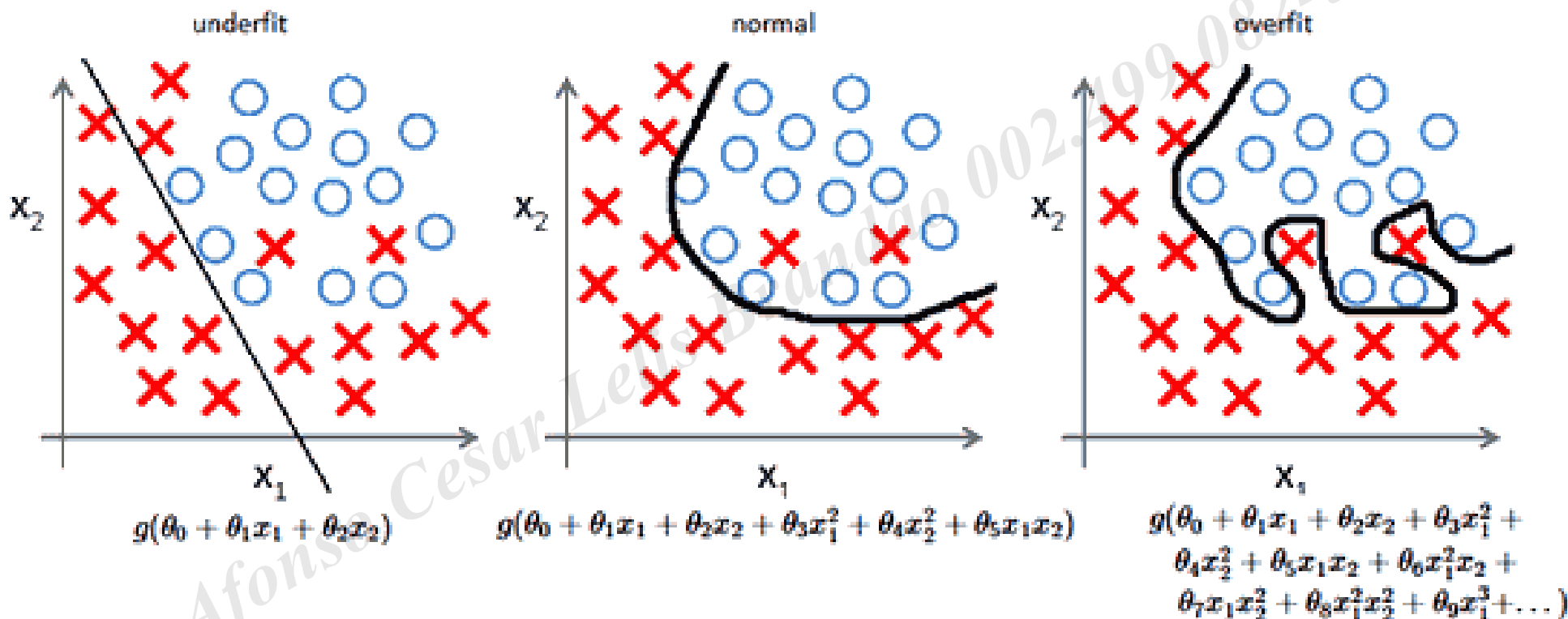
1. Minimum number of observations per sheet
2. Maximum depth
3. Cost of complexity

Cost of complexity



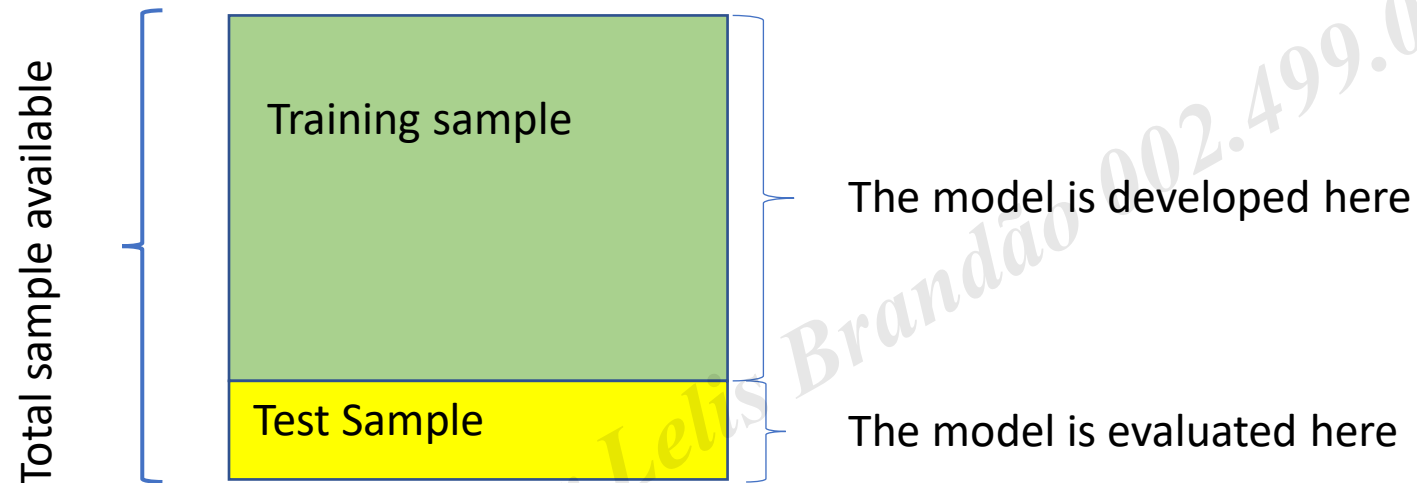
Cost of complexity	Low	Moderate	High
	High	Mean	Low

Over fitting / Under fitting



<http://mlwiki.org/index.php/Overfitting>

Cross validation



The most simple strategy is to split the basis into training and test. We develop the model in the training base, and evaluate the test base.

Decision Trees:

Where do they live?

What are they?

What do they eat?

Natural Predators



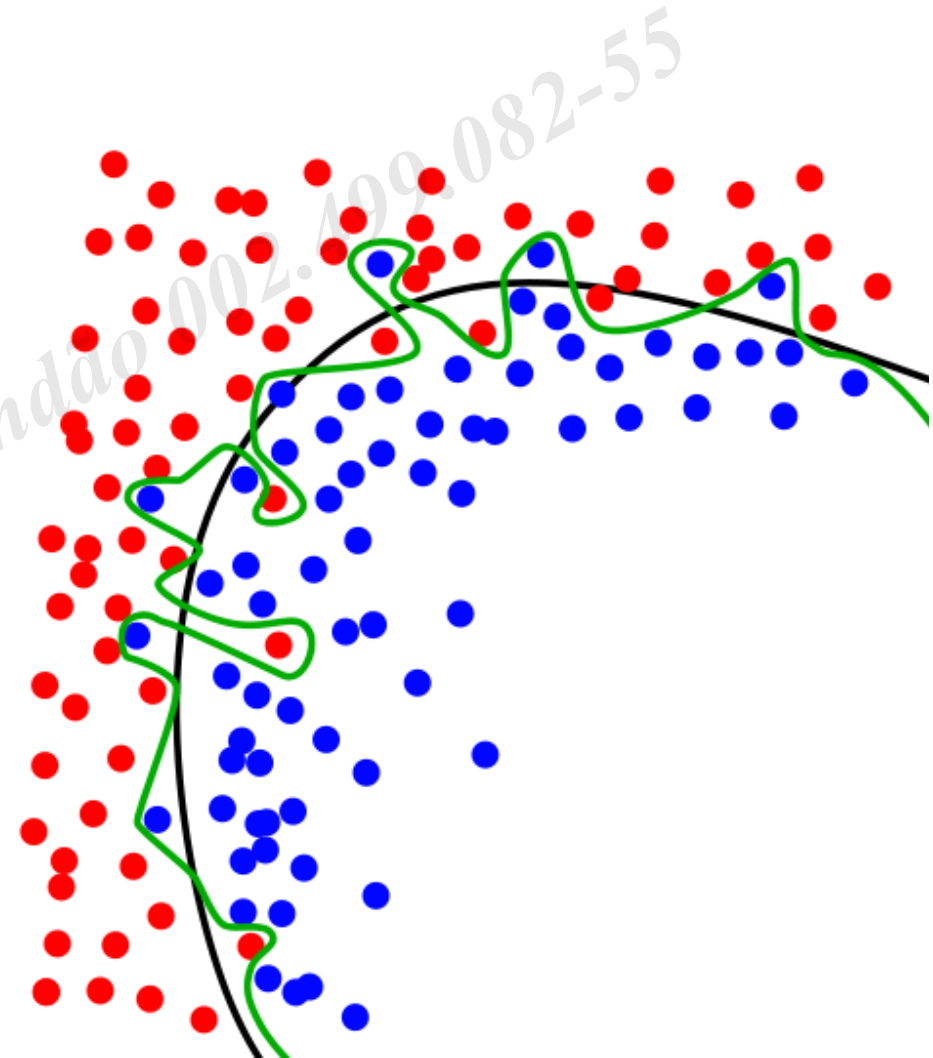
THE BEST WAY TO EXPLAIN OVERFITTING

What is it?

How to avoid it?

Overfitting

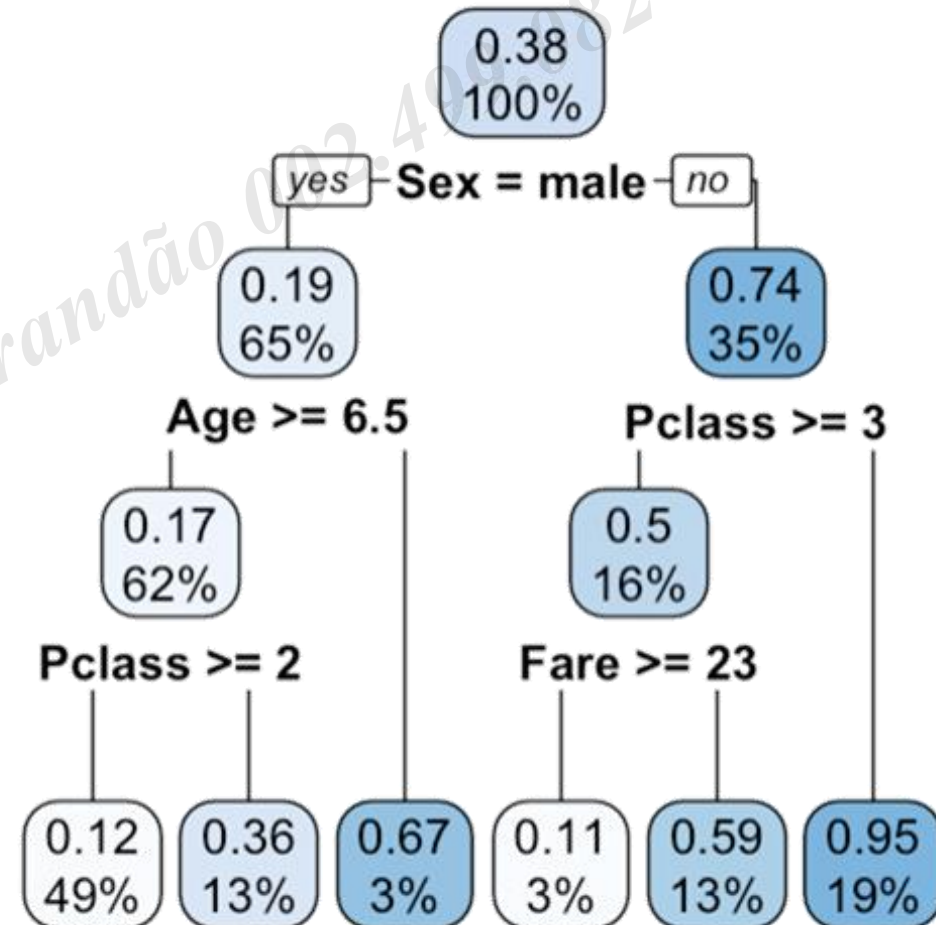
- *Overfitting* is the phenomenon of the model adjusting to very particular characteristics of the sample that will not be repeated in other samples.
- This implies that the quality of the model may be much lower in the application of the model.



The tree as a classifier

Requisites:

To have all variables.



The tree as a classifier

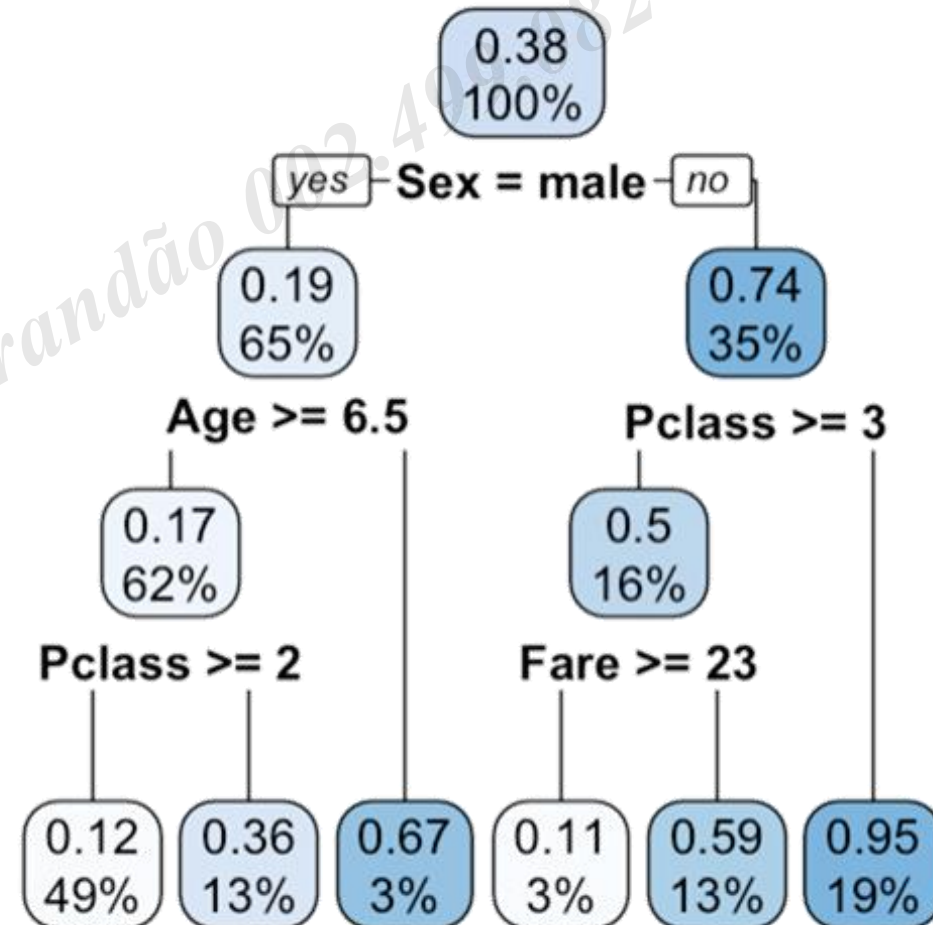
Probability of the F. event:

$$P(S|F) = \frac{N_f^S}{N_f}$$

$P(S|F)$ - probability of success of the F sheet

N_f - it is the number of individuals on F sheet

N_f^S - it is the number of survivors on F sheet



The tree as a classifier

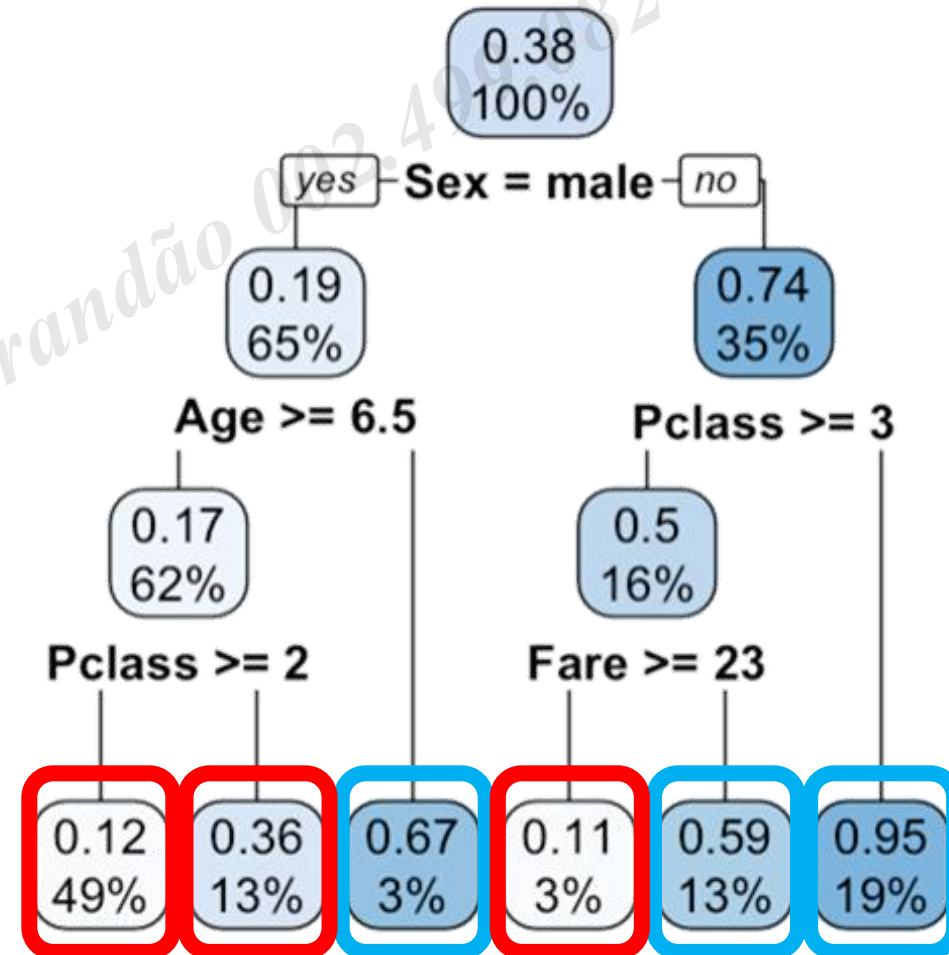
Classification:

Standard classification:

Survivor: $P(S|F) \geq 50\% \Rightarrow C(F) = "Y"$

No survivors: $P(S|F) < 50\% \Rightarrow C(F) = "N"$

Predicted value	True Value	
	0	1
0	484	96
1	65	246



Evaluation of the model

- Accuracy:

Hits on attempts

Predicted value	True Value	
	0	1
0	484	96
1	65	246

In this example:

$$\frac{484 + 246}{891} = 82\%$$

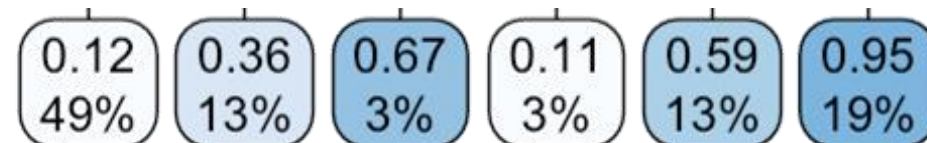
Trees as diagnosis

$$\text{Sensitivity: } \frac{TP}{FN+TP} = \frac{246}{246+96} = 72\%$$

$$\text{Specificity: } \frac{TN}{TN+FP} = \frac{484}{484+65} = 72\%$$

Predicted value	True Value	
	0	1
0	484	96
1	65	246

Predicted value	True Value	
	0	1
0	TN	FN
1	FP	TP



Diagnosis and cutoff points

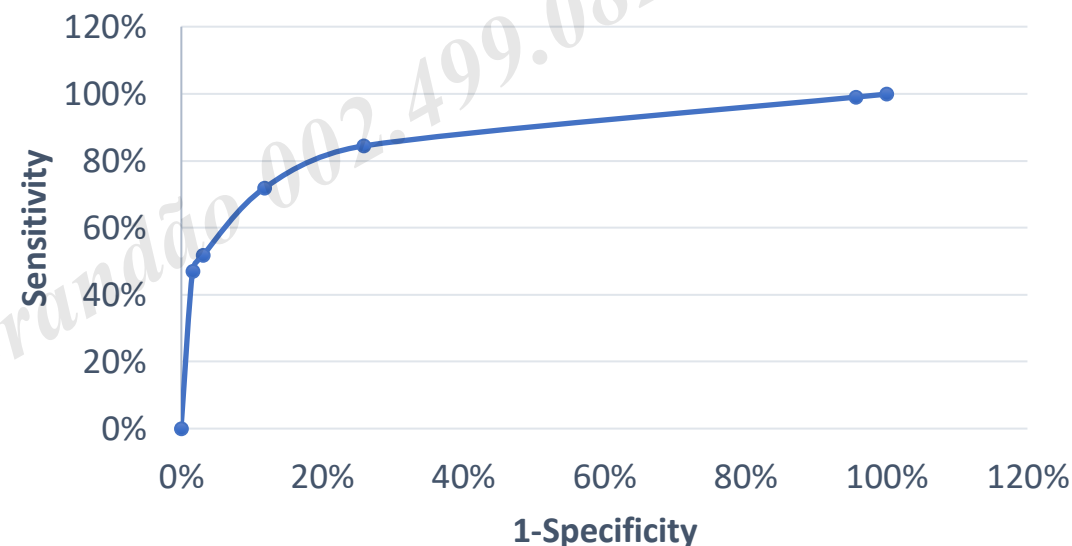
CUT	TP	FP	TN	FN
0% - 11.1%	342	549	0	0
11.1% - 11.5%	339	525	24	3
11.5% - 35.8%	289	142	407	53
35.8% - 58.9%	246	65	484	96
58.9% - 66.7%	177	17	532	165
66.7% - 94.7%	161	9	540	181
94.7% - 100%	0	0	549	342

Accuracy	Specificity	1-Specificity	Sensibility
38%	0%	100%	100%
41%	4%	96%	99%
78%	74%	26%	85%
82%	88%	12%	72%
80%	97%	3%	52%
79%	98%	2%	47%
62%	100%	0%	0%

For each cutoff point, we have a confusion matrix.
In this case, we have 8 possible matrices with the trained tree.

ROC Curve

CUT	1-Specificity	Sensibility
0% - 11.1%	100%	100%
11.1% - 11.5%	96%	99%
11.5% - 35.8%	26%	85%
35.8% - 58.9%	12%	72%
58.9% - 66.7%	3%	52%
66.7% - 94.7%	2%	47%
94.7% - 100%	0%	0%



ROC curve is a graphical plot of 1-Specificity on the X-axis by Sensitivity on the Y-axis, obtained for each possible cutoff point of the classifier.



OMML1 _script02-Algoritmo_avaliacao_overfitting

Homework

script03



OMML1_script03_Exercício for home



CONCLUSION

- Robust, interpretable, flexible
- Without probabilistic assumptions
- It's necessary *cross-validation*

Quanto mais aprendo, mais
tenho certeza de que, o que
sei, é apenas uma gota,
diante do oceano do que
ainda preciso aprender.



PENSADOR

Jose Ap Barcelos



That's it for
today ;)



[linkedin.com/in/joao-serrajordia](https://www.linkedin.com/in/joao-serrajordia)

Famous Algorithms

- CART
- CHAID
- ID3
- C4.5
- C5.0

Interesting stack overflow on this:

<https://stackoverflow.com/questions/9979461/different-decision-tree-algorithms-with-comparison-of-complexity-or-performance>

In the next class...

Techniques of cross-validation

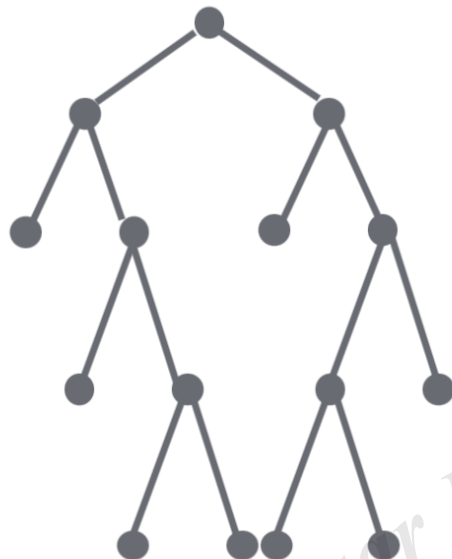


Extras

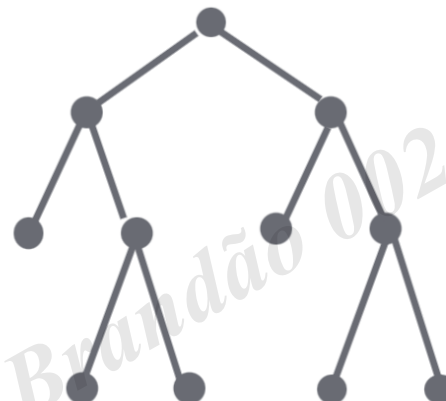


OMML1 _script02-Algoritmo_avalicao_overfitting

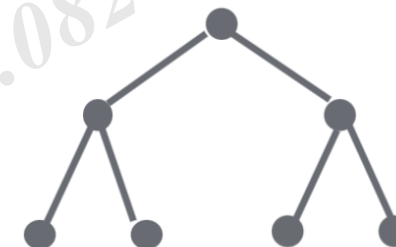
Pruning



Base de treino: 95%
Base de validação: 40%



Base de treino: 70%
Base de validação: 60%



Base de treino: 65%
Base de validação: 64%

Acurácia

Amostra de treino

Amostra de validação

Cross validation strategies

Escolher parâmetros do modelo com uma base de validação ainda pode propiciar overfitting.

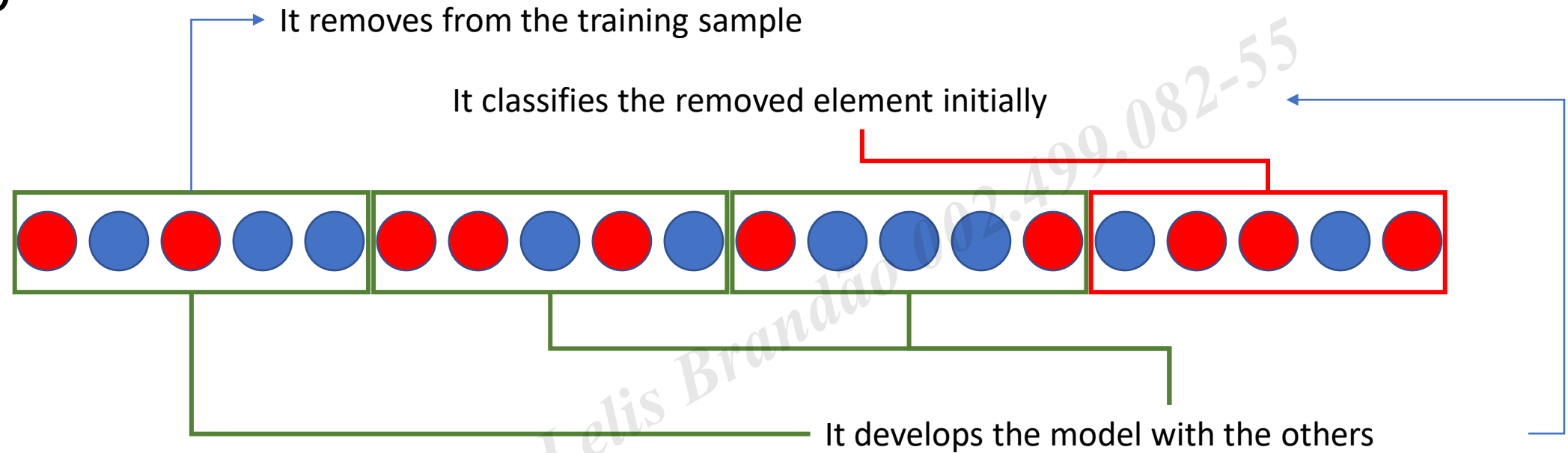
Há diversas técnicas de validação cruzada para se evitar esse efeito. No momento vou mencionar uma técnica clássica: dividir a base em Treino, Validação e Teste

+
Amostra de treino

+
Amostra de validação

+
Amostra de teste

K-fold



- We divide the base into sub-samples k
- For each sub-sample:
 - We remove the sub-sample as validation
 - We train the model with the remaining observations
 - We use this model to classify the removed sub-sample
 - We evaluate the metrics of the model's performance
- We calculate the average of the metrics of the model's performance

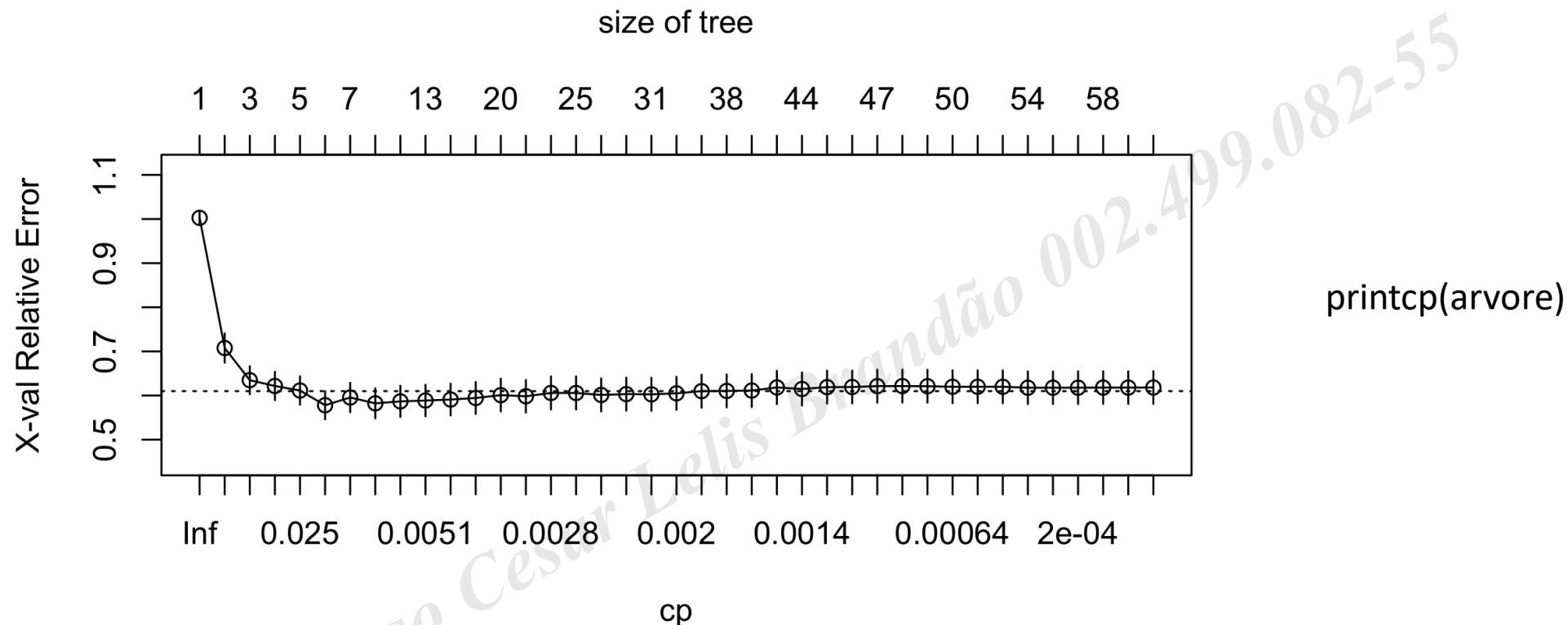
K-fold

Tipicamente, fazemos o mesmo para variações do modelo para otimizar hiperparâmetros.



	Acurácia 1	Acurácia 2	Acurácia 3	Acurácia 4	Acurácia Média
Modelo 1	62%	58%	61%	59%	60%
Modelo 2	50%	51%	49%	47%	49%
Modelo 3	72%	68%	71%	75%	72%

Post-pruning with cross validation



R performs the pruning of the tree by performing a k -fold to optimize the CP (complexity path), a parameter that summarizes the complexity of the tree. This is made with a k -fold.

THANK YOU!

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